

BÀI TẬP CHƯƠNG I & II

Bài 1.1 → 1.2 (Vở ghi lý thuyết)

Bài 1.3:

$$A = \begin{pmatrix} 1 & -4 & 2 \\ 1 & -4 & 2 \\ 1 & -4 & 2 \end{pmatrix}$$

$$\begin{aligned} A^2 &= A \cdot A = \begin{pmatrix} 1 & -4 & 2 \\ 1 & -4 & 2 \\ 1 & -4 & 2 \end{pmatrix} \begin{pmatrix} 1 & -4 & 2 \\ 1 & -4 & 2 \\ 1 & -4 & 2 \end{pmatrix} \\ &= \begin{pmatrix} -1 & 4 & -2 \\ -1 & 4 & -2 \\ -1 & 4 & -2 \end{pmatrix} = -A \end{aligned}$$

$$A^3 = A^2 \cdot A = (-A) \cdot A = -A^2 = A$$

$$A^4 = A^3 \cdot A = A \cdot A = A^2 = -A$$

Vậy ta có:

$$A^n = \begin{cases} A & \text{nếu } n = 2k+1 \\ -A & \text{nếu } n = 2k \end{cases}$$

$$\Rightarrow A^{200} + A = (-A) + A = O$$

Bài 1.4:

$$A = \begin{pmatrix} -10 & 11 & -22 \\ 3 & -2 & 6 \\ 6 & -6 & 13 \end{pmatrix}$$

$$A^2 = A \cdot A = \begin{pmatrix} -10 & 11 & -22 \\ 3 & -2 & 6 \\ 6 & -6 & 13 \end{pmatrix} \begin{pmatrix} -10 & 11 & -22 \\ 3 & -2 & 6 \\ 6 & -6 & 13 \end{pmatrix}$$

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1. $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$

$A^{2018} = (A^2)^{1009} = I^{1009} = I$

$A^{2019} = A^{2018} A = I A = A$

b. $\det(A^{2018} + 3A^n)$

Có $A^n = \begin{cases} I \text{ nếu } n = 2k \text{ (chẵn)} \\ A \text{ nếu } n = 2k+1 \text{ (lẻ)} \end{cases}$

Nếu $n = 2k$ thì $A^n = I$

$\rightarrow A^{2018} + 3A^n = I + 3I = 4I$

$\det(A^{2018} + 3A^n) = \det(4I) = \begin{vmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{vmatrix} = 64$

Nếu n lẻ thì $A^n = A$

$\rightarrow A^{2018} + 3A^n = I + 3A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -30 & 33 & -6 \\ 9 & -6 & 18 \\ 18 & -18 & 40 \end{pmatrix}$

$= \begin{pmatrix} -29 & 33 & -6 \\ 9 & -5 & 18 \\ 18 & -18 & 40 \end{pmatrix}$

$\det(A^{2018} + 3A^n) = \det(I + 3A)$

$= \begin{vmatrix} -29 & 33 & -6 \\ 9 & -5 & 18 \\ 18 & -18 & 40 \end{vmatrix} = -66$

$= \begin{vmatrix} 9 & -5 & 18 \\ 18 & -18 & 40 \\ -29 & 33 & -6 \end{vmatrix} = -32$

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2. $A = \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & -1 \end{pmatrix}$

$A^2 = A \cdot A = \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & -1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$

$A^{2018} = (A^2)^{1009} = I^{1009} = I$

$A^{2019} = (A^2)^{1009} A = A$

b. $\det(SA^m + 7A^n)$

Có $A^n = \begin{cases} I \text{ nếu } n \text{ chẵn} \\ A \text{ nếu } n \text{ lẻ} \end{cases}$

* Nếu n chẵn $A^n = I$, n lẻ $A^n = A$

$\rightarrow SA^m + 7A^n = 5I + 7I = 12I$

$\det(SA^m + 7A^n) = \det(12I)$

$= \begin{vmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{vmatrix} = 1728$

* Nếu n chẵn $A^n = I$, n lẻ $A^n = A$

$\rightarrow SA^m + 7A^n = 5I + 7A$

$= \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} + 7 \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & -1 \end{pmatrix}$

$= \begin{pmatrix} 12 & 7a & 7a \\ 7b & 12 & 7b \\ 7a & 7b & -2 \end{pmatrix}$

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$$= \begin{pmatrix} 12 & 0 & 79 \\ 0 & 12 & 76 \\ 0 & 0 & -2 \end{pmatrix}$$

det. $(SA^m + 7A^n) = \det(SA + 7A)$

* Nếu m < n, $A^m = A$, n chẵn $A^n = A \cdot I$
 $\rightarrow SA^m + 7A^n = SA + 7I$

$$= \begin{pmatrix} 5 & 0 & 50 \\ 0 & 5 & 56 \\ 0 & 0 & -5 \end{pmatrix} + \begin{pmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{pmatrix}$$

$$= \begin{pmatrix} 12 & 0 & 50 \\ 0 & 12 & 56 \\ 0 & 0 & 12 \end{pmatrix} = -288$$

* Nếu m < n, $A^m = A$, n lẻ, $A^n = A$
 $\rightarrow SA^m + 7A^n = SA + 7A = 12A$

$$= \begin{pmatrix} 12 & 0 & 129 \\ 0 & 12 & 129 \\ 0 & 0 & -129 \end{pmatrix}$$

det. $(SA^m + 7A^n) = \det(12A) = 12 \cdot 12 \cdot (-12) = -1728$

Bài 16 $\rightarrow$ 118 (lưu ý trong vở lý thuyết)

Bài 19:

$$A = \begin{pmatrix} 1 & 3 & -2 \\ 2 & 1 & 3 \\ 5 & 4 & 7 \end{pmatrix}$$

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0) $A^2 = \begin{pmatrix} 1 & 3 & -2 \\ 2 & 1 & 3 \\ 5 & 4 & 7 \end{pmatrix} \begin{pmatrix} 1 & 3 & -2 \\ 2 & 1 & 3 \\ 5 & 4 & 7 \end{pmatrix}$

$$= \begin{pmatrix} -3 & -2 & -7 \\ 10 & 19 & 20 \\ 48 & 47 & 51 \end{pmatrix}$$

$A^3 = A^2 \cdot A = \begin{pmatrix} -3 & -2 & -7 \\ 10 & 19 & 20 \\ 48 & 47 & 51 \end{pmatrix} \cdot \begin{pmatrix} 1 & 3 & -2 \\ 2 & 1 & 3 \\ 5 & 4 & 7 \end{pmatrix}$

$$= \begin{pmatrix} -42 & -39 & -49 \\ 157 & 156 & 159 \\ 397 & 395 & 402 \end{pmatrix}$$

$A^4 = A^3 \cdot A = \begin{pmatrix} -365 & -361 & -376 \\ 1269 & 1263 & 1267 \\ 3197 & 3194 & 3205 \end{pmatrix}$

$\rightarrow \det(A^4 + 3A^3) = -61952$

VD: $D = \begin{vmatrix} 2 & x & x & x \\ x & 2 & x & x \\ x & x & 2 & x \\ x & x & x & 2 \end{vmatrix}$

$H_1 + H_2 + H_3 + H_4 \rightarrow H_1$

$$= \begin{vmatrix} 3x+2 & 3x+2 & 3x+2 & 3x+2 \\ x & 2 & x & x \\ x & x & 2 & x \\ x & x & x & 2 \end{vmatrix}$$

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$H_2 - 2H_1 \rightarrow H_2$ (3×12) $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 2-2 & 0 & 0 \\ 0 & 0 & 2-2 & 0 \\ 0 & 0 & 0 & 2-2 \end{pmatrix}$

$H_3 - 2H_1 \rightarrow H_3$

$H_4 - 2H_1 - 2H_2$

$(12+22)(2-2)^3$

$D=0 \Rightarrow \begin{cases} 3x+2=0 \\ (2-x)^3=0 \end{cases} \Rightarrow \begin{cases} x=-2 \\ x=-2/3 \end{cases}$

b) $A+5I = \begin{pmatrix} 1 & 3 & -2 \\ 2 & 1 & 3 \\ 5 & 4 & 7 \end{pmatrix} + 5 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 6 & 3 & -2 \\ 2 & 6 & 3 \\ 5 & 4 & 12 \end{pmatrix} \times 2$

$\det(5I+A) = 377 \neq 0$

$\text{Veg } \lambda(5I+A) = 3$

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$A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 1 & 3 \\ 3 & 4 & 4 \end{pmatrix}$ $B = \begin{pmatrix} 1 & 2 \\ -1 & 3 \\ 3 & 4 \end{pmatrix}$

a) $\det(A^3 B^2 + 4A^2 B^3)$ $\cancel{80}$

a) $\det(AB)$

$A \cdot B = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 1 & 3 \end{pmatrix} \times 2 \times 3$ $\begin{pmatrix} 1 & 2 \\ -1 & 3 \\ 3 & 4 \end{pmatrix} \times 3 \times 2$ $\begin{pmatrix} 1 & 2 \\ -1 & 3 \\ 3 & 4 \end{pmatrix}$ $\times 3 \times 2$ $\begin{pmatrix} 1 & 2 \\ -1 & 3 \\ 3 & 4 \end{pmatrix}$

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$= \begin{pmatrix} 8 & 20 \\ 7 & 13 \end{pmatrix}$

$\det(AB) = \begin{vmatrix} 8 & 20 \\ 7 & 13 \end{vmatrix} = -36$

a) $\det(BA)$

$BA = \begin{pmatrix} 1 & 2 \\ -1 & 3 \\ 3 & 4 \end{pmatrix} \times 2 \times 3$ $\begin{pmatrix} 1 & 2 & 3 \\ -1 & 1 & 3 \\ -1 & 1 & 3 \end{pmatrix}$

$= \begin{pmatrix} -1 & 4 & 9 \\ -9 & 1 & 6 \\ -1 & 10 & 21 \end{pmatrix}$

$\det(BA) = \begin{vmatrix} -1 & 4 & 9 \\ -9 & 1 & 6 \\ -1 & 10 & 21 \end{vmatrix}$

$= (-1) \cdot 1 \cdot 21 + (-4) \cdot 10 \cdot 9 + 4 \cdot 6 \cdot (-1)$

$= (-1) \cdot 21 - 4 \cdot 90 - 24$

$= 0$

b) $\det(BA + 4I)$

$BA + 4I = \begin{pmatrix} -1 & 4 & 9 \\ -9 & 1 & 6 \\ -1 & 10 & 21 \end{pmatrix} + 4 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 3 & 4 & 9 \\ -9 & 5 & 6 \\ -1 & 10 & 25 \end{pmatrix}$

$\det(BA + 4I) = \begin{vmatrix} 3 & 4 & 9 \\ -9 & 5 & 6 \\ -1 & 10 & 25 \end{vmatrix} = 256 \cdot 20$

$\rightarrow \det(BA + 4I) = 3$

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$\det(BA + 4I) = 256 \cdot 20$

$\rightarrow \det(BA + 4I) = 3$

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Bài 1.11

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 3 & 1 \\ 2 & 3 \end{pmatrix}$$

a) $\det(A^2 B^2 + 4A^2 B^2) = A^2(A+4B)B^2$

$\det(A^2 B^2 + 4A^2 B^2) = \det A^2 \det(A+4B) \det B^2$

$A^2 = A \cdot A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 18 & 14 \\ 7 & 11 \end{pmatrix}$

$\det A^2 = \begin{vmatrix} 18 & 14 \\ 7 & 11 \end{vmatrix} = 100$

$\det B^2 = 49$

$A+4B = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} + \begin{pmatrix} 12 & 4 \\ 8 & 12 \end{pmatrix} = \begin{pmatrix} 16 & 6 \\ 9 & 15 \end{pmatrix}$

$= \begin{vmatrix} 16 & 6 \\ 9 & 15 \end{vmatrix} = 186$

$\det(A+4B) = \begin{vmatrix} 16 & 6 \\ 9 & 15 \end{vmatrix} = 186$

$\text{Vậy } \det(A^2 B^2 + 4A^2 B^2) = 100 \cdot 49 \cdot 186 = 911400$

Bài 1.12

$A = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & -2 \\ 3 & -2 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 3 & 4 & 5 \\ 2 & 2 & 3 \\ 1 & -1 & 3 \end{pmatrix}$

$A \cdot B = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & -2 \\ 3 & -2 & 1 \end{pmatrix} \begin{pmatrix} 3 & 4 & 5 \\ 2 & 2 & 3 \\ 1 & -1 & 3 \end{pmatrix}$

$= \begin{pmatrix} 18 & 14 & 10 \\ 7 & 11 & 11 \\ 10 & 9 & 10 \end{pmatrix}$

$\det(A \cdot B) = \begin{vmatrix} 18 & 14 & 10 \\ 7 & 11 & 11 \\ 10 & 9 & 10 \end{vmatrix} = 101$

$\det A = \begin{vmatrix} 1 & 2 & 4 \\ 2 & 1 & -2 \\ 3 & -2 & 1 \end{vmatrix} = 101$

$\det B = \begin{vmatrix} 3 & 4 & 5 \\ 2 & 2 & 3 \\ 1 & -1 & 3 \end{vmatrix} = 101$

$\det(A \cdot B) = \det A \cdot \det B = 101 \cdot 101 = 10201$

Bài 1.13

$$= \begin{pmatrix} 23 & 4 & 13 \\ 0 & 12 & 7 \\ 9 & 7 & 12 \end{pmatrix}$$

$\det(A \cdot B) = \begin{vmatrix} 23 & 4 & 13 \\ 0 & 12 & 7 \\ 9 & 7 & 12 \end{vmatrix} = -47$

b) $(A+2B)^2 - 19(A+2B)$

$A+2B = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} + 2 \begin{pmatrix} 3 & 1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 10 & 4 \\ 5 & 9 \end{pmatrix}$

$(A+2B)^2 = \begin{pmatrix} 10 & 4 \\ 5 & 9 \end{pmatrix} \begin{pmatrix} 10 & 4 \\ 5 & 9 \end{pmatrix} = \begin{pmatrix} 120 & 76 \\ 95 & 101 \end{pmatrix}$

$(A+2B)^2 - 19(A+2B) = \begin{pmatrix} 120 & 76 \\ 95 & 101 \end{pmatrix} - 19 \begin{pmatrix} 10 & 4 \\ 5 & 9 \end{pmatrix} = \begin{pmatrix} -70 & 0 \\ 0 & -70 \end{pmatrix}$

$= \begin{pmatrix} -70 & 0 \\ 0 & -70 \end{pmatrix} = -70D$

Bài 1.13

$A = \begin{pmatrix} 3 & 5 & 7 \\ 2 & 3 & 2 \\ 2 & 2 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 4 & -5 \\ -2 & 2 & 3 \\ 4 & -1 & 2 \end{pmatrix}$

$\det(A^2 B - 3AB^2) = A(A-3B) \det B$

$A^2 B - 3AB^2 = \det A \cdot \det(A-3B) \cdot \det B$

$\det A = \begin{vmatrix} 3 & 5 & 7 \\ 2 & 3 & 2 \\ 2 & 2 & 3 \end{vmatrix} = 105$

$\det(A-3B) = \begin{vmatrix} 1 & 4 & -5 \\ -2 & 2 & 3 \\ 4 & -1 & 2 \end{vmatrix} = 105$

$\det B = \begin{vmatrix} 1 & 4 & -5 \\ -2 & 2 & 3 \\ 4 & -1 & 2 \end{vmatrix} = 101$

$\det(A^2 B - 3AB^2) = 105 \cdot 105 \cdot 101 = 1070505$

$\det B = \begin{vmatrix} 1 & 4 & -5 \\ -2 & 2 & 3 \\ 4 & -1 & 2 \end{vmatrix} = 101$

$\det(A \cdot B) = \det A \cdot \det B = 105 \cdot 101 = 10605$

$\det(A \cdot B) = 10605$

$$A - 3B = \begin{pmatrix} 3 & 5 & 7 \\ 2 & 3 & -2 \\ 2 & 2 & 3 \end{pmatrix} - 3 \begin{pmatrix} 1 & 4 & -5 \\ -2 & 2 & 3 \\ 4 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -7 & 22 \\ 8 & -3 & -14 \\ -10 & 1 & -3 \end{pmatrix}$$

$$\det(A - 3B) = \begin{vmatrix} 0 & -7 & 22 \\ 8 & -3 & -14 \\ -10 & 1 & -3 \end{vmatrix} = -1422$$

$$\text{Veg det}(A^2B - 3AB^2) = (-1422) \cdot (-105) = 149310$$

Bài 1.14.

$$A = \begin{pmatrix} 1 & 2 & 5 \\ 3 & 4 & -1 \\ 4 & 2 & -3 \end{pmatrix} \quad B = \begin{pmatrix} 3 & 2 & -2 \\ 3 & 1 & 4 \\ 5 & 2 & 7 \end{pmatrix}$$

a) $\det(A^3B^2 - 3A^2B^3) = A^2(A - 3B)B^2$

$$\det(A^3B^2 - 3A^2B^3) = \det A^2 \det(A - 3B) \det B^2$$

$$A^2 = A \cdot A = \begin{pmatrix} 1 & 2 & 5 \\ 3 & 4 & -1 \\ 4 & 2 & -3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 5 \\ 3 & 4 & -1 \\ 4 & 2 & -3 \end{pmatrix} = \begin{pmatrix} 27 & 20 & -12 \\ -11 & 20 & 14 \\ -2 & -16 & 27 \end{pmatrix}$$

$$\det A^2 = 2500$$

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$$B^2 = B \cdot B = \begin{pmatrix} 3 & 2 & -2 \\ 3 & 1 & 4 \\ 5 & 2 & 7 \end{pmatrix} \begin{pmatrix} 3 & 2 & -2 \\ 3 & 1 & 4 \\ 5 & 2 & 7 \end{pmatrix} = \begin{pmatrix} 5 & 4 & -12 \\ 32 & 15 & 26 \\ 58 & 26 & 47 \end{pmatrix}$$

$$\det B^2 = \begin{vmatrix} 5 & 4 & -12 \\ 32 & 15 & 26 \\ 58 & 26 & 47 \end{vmatrix} = 49$$

$$A - 3B = \begin{pmatrix} 1 & 2 & 5 \\ 3 & 4 & -1 \\ 4 & 2 & -3 \end{pmatrix} - 3 \begin{pmatrix} 3 & 2 & -2 \\ 3 & 1 & 4 \\ 5 & 2 & 7 \end{pmatrix} = \begin{pmatrix} -8 & -4 & 11 \\ -6 & 1 & -13 \\ -11 & -4 & -24 \end{pmatrix}$$

$$\det(A - 3B) = \begin{vmatrix} -8 & -4 & 11 \\ -6 & 1 & -13 \\ -11 & -4 & -24 \end{vmatrix} = 10976$$

$$\det(A^3B^2 - 3A^2B^3) = 2500 \cdot 49 \cdot 10976 = 122432500$$

~~10045~~ b) $\sim (A + 3B)$

$$A + 3B = \begin{pmatrix} 1 & 2 & 5 \\ 3 & 4 & -1 \\ 4 & 2 & -3 \end{pmatrix} + 3 \begin{pmatrix} 3 & 2 & -2 \\ 3 & 1 & 4 \\ 5 & 2 & 7 \end{pmatrix} = \begin{pmatrix} 10 & 8 & -1 \\ 12 & 7 & 11 \\ 19 & 8 & 18 \end{pmatrix}$$

$$\det(A + 3B) = \begin{vmatrix} 10 & 8 & -1 \\ 12 & 7 & 11 \\ 19 & 8 & 18 \end{vmatrix} = 321 \neq 0$$

$$\rightarrow \lambda(A + 3B) = 3$$

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Đặt 1.15:

$$A = \begin{pmatrix} 3 & 2 & -2 \\ 1 & 1 & 3 \\ 2 & -2 & 1 \end{pmatrix} \quad B = \begin{pmatrix} -2 & 4 & 5 \\ 1 & 2 & -3 \\ 2 & -1 & -1 \end{pmatrix}$$

g) $A^3 B^2 + 2A^2 B^3$ Cho nhiều $\in \mathbb{R}$ $\det(A^3 B^2 + 2A^2 B^3) \neq 0$

(c) $A^3 B^2 + 2A^2 B^3 = A^2(A + 3B)B^2$

$\rightarrow \det(A^3 B^2 + 2A^2 B^3) = \det A^2 \cdot \det(A + 3B) \cdot \det B^2$

$$A^2 = \begin{pmatrix} 3 & 2 & -2 \\ 1 & 1 & 3 \\ 2 & -2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 & -2 \\ 1 & 1 & 3 \\ 2 & -2 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 12 & -2 \\ 10 & -3 & 4 \\ 6 & 0 & -9 \end{pmatrix}$$

$$\det A^2 = \begin{vmatrix} 7 & 12 & -2 \\ 10 & -3 & 4 \\ 6 & 0 & -9 \end{vmatrix} = 1521$$

$$B^2 = \begin{pmatrix} -2 & 4 & 5 \\ 1 & 2 & -3 \\ 2 & -1 & -1 \end{pmatrix} \cdot \begin{pmatrix} -2 & 4 & 5 \\ 1 & 2 & -3 \\ 2 & -1 & -1 \end{pmatrix} = \begin{pmatrix} -16 & -5 & -14 \\ -6 & 11 & 4 \\ -3 & 5 & 14 \end{pmatrix}$$

$$\det B^2 = 2601$$

$$A + 3B = \begin{pmatrix} 3 & 2 & -2 \\ 1 & 1 & 3 \\ 2 & -2 & 1 \end{pmatrix} + 3 \begin{pmatrix} -2 & 4 & 5 \\ 1 & 2 & -3 \\ 2 & -1 & -1 \end{pmatrix} = \begin{pmatrix} -3 & 16 & 13 \\ 4 & 7 & 6 \\ 8 & -5 & 4 \end{pmatrix}$$

$$\det(A + 3B) = \begin{vmatrix} -3 & 16 & 13 \\ 4 & 7 & 6 \\ 8 & -5 & 4 \end{vmatrix} = -650$$

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$$\det(A^2 B^3 + 3A^2 B^3) = -650 \cdot 1521 \cdot \det A = \neq 0$$

b) $\det(A^2 B - 2AB^2)$

(c) $A^2 B - 2AB^2 = A(A - 2B)B = D$

$$A - 2B = \begin{pmatrix} 3 & 2 & -2 \\ 1 & 1 & 3 \\ 2 & -2 & 1 \end{pmatrix} - 2 \begin{pmatrix} -2 & 4 & 5 \\ 1 & 2 & -3 \\ 2 & -1 & -1 \end{pmatrix} = \begin{pmatrix} 7 & -6 & -12 \\ -1 & -3 & 9 \\ -2 & 0 & -1 \end{pmatrix}$$

$$\det(A - 2B) = 201$$

$$\det A = \begin{vmatrix} 3 & 2 & -2 \\ 1 & 1 & 3 \\ 2 & -2 & 1 \end{vmatrix} = 156$$

$$\det B = \begin{vmatrix} -2 & 4 & 5 \\ 1 & 2 & -1 \\ 2 & -1 & 1 \end{vmatrix} = -51$$

$$\det(D) = 201 \cdot 156 \cdot (-51) = \neq 0$$

Đặt 1.16:

$$A = \begin{pmatrix} 2 & 1 & -3 \\ 1 & 3 & -2 \\ -1 & 2 & 1 \end{pmatrix} \quad A^{-1} = ?$$

$$\det A = \begin{vmatrix} 2 & 1 & -3 \\ 1 & 3 & -2 \\ -1 & 2 & 1 \end{vmatrix} = -9 \neq 0$$

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Việc cần phải làm đầu tiên

$$A_{11} = \begin{vmatrix} 3 & -2 \\ -2 & 1 \end{vmatrix} = -1$$

$$A_{12} = \begin{vmatrix} 1 & -2 \\ -1 & 1 \end{vmatrix} = 1$$

$$A_{13} = \begin{vmatrix} 1 & 1 \\ -1 & -2 \end{vmatrix} = 1$$

$$A_{21} = 5$$

$$A_{22} = -1$$

$$A_{23} = 3$$

Việc tiếp theo

$$A^* = \begin{pmatrix} -1 & 1 & 1 \\ 5 & -1 & 3 \\ 7 & 1 & 5 \end{pmatrix}^T = \begin{pmatrix} -1 & 5 & 7 \\ 1 & -1 & 1 \\ 1 & 3 & 5 \end{pmatrix}$$

NT yêu cầu

$$A^{-1} = \frac{1}{\det A} A^* = \frac{-1}{4} \begin{pmatrix} -1 & 5 & 7 \\ 1 & -1 & 1 \\ 1 & 3 & 5 \end{pmatrix}$$

Việc tiếp theo

$$A = \begin{pmatrix} 2 & 0 & 3 \\ 1 & 2 & 2 \\ 1 & 0 & 4 \end{pmatrix}$$

4) $A^3 - 8A^2 + 17A = 2$

$$A^1 = \begin{pmatrix} 2 & 0 & 3 \\ 1 & 2 & 2 \\ 1 & 0 & 4 \end{pmatrix} \begin{pmatrix} 2 & 0 & 3 \\ 1 & 2 & 2 \\ 1 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 0 & 18 \\ 6 & 4 & 15 \\ 6 & 4 & 15 \end{pmatrix}$$

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$$A^3 = \begin{pmatrix} 7 & 0 & 18 \\ 6 & 4 & 15 \\ 6 & 0 & 19 \end{pmatrix} \begin{pmatrix} 2 & 0 & 3 \\ 1 & 2 & 2 \\ 1 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 32 & 0 & 93 \\ 34 & 8 & 86 \\ 34 & 0 & 94 \end{pmatrix}$$

$$A^3 - 8A^2 + 17A = \begin{pmatrix} 32 & 0 & 93 \\ 34 & 8 & 86 \\ 34 & 0 & 94 \end{pmatrix} - 8 \begin{pmatrix} 7 & 0 & 18 \\ 6 & 4 & 15 \\ 6 & 0 & 19 \end{pmatrix} + 17 \begin{pmatrix} 2 & 0 & 3 \\ 1 & 2 & 2 \\ 1 & 0 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 10 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 10 \end{pmatrix} = 10I$$

b) $\det A = \begin{vmatrix} 2 & 0 & 3 \\ 1 & 2 & 2 \\ 1 & 0 & 4 \end{vmatrix} = 10 \neq 0$

Việc cần phải làm tiếp

$$A_{11} = \begin{vmatrix} 2 & 2 \\ 0 & 4 \end{vmatrix} = 8$$

$$A_{12} = \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = -2$$

$$A_{13} = \begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} = -2$$

$$A_{21} = \begin{vmatrix} 0 & 3 \\ 0 & 4 \end{vmatrix} = 0$$

$$A_{22} = \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} = 5$$

$$A_{23} = \begin{vmatrix} 2 & 0 \\ 1 & 0 \end{vmatrix} = 0$$

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Tính các phần bù đại số

$$A_{11} = \begin{vmatrix} 1 & -1 \\ -2 & 1 \end{vmatrix} = -1 \quad A_{21} = \begin{vmatrix} -1 & 4 \\ -2 & 1 \end{vmatrix} = -7$$

$$A_{12} = \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} = -3 \quad A_{22} = \begin{vmatrix} 1 & -4 \\ 1 & 1 \end{vmatrix} = -3$$

$$A_{13} = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = -5 \quad A_{23} = \begin{vmatrix} -1 & -1 \\ 1 & -2 \end{vmatrix} = -1$$

$$A_{31} = \begin{vmatrix} -1 & 4 \\ 1 & -1 \end{vmatrix} = -3 \quad A_{32} = \begin{vmatrix} -1 & 4 \\ 2 & -1 \end{vmatrix} = 9$$

$$A_{33} = \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = -3$$

NT phụ trợ:

$$A^* = \begin{pmatrix} -1 & -3 & -5 \\ -7 & -3 & 1 \\ -3 & 9 & 3 \end{pmatrix} \quad T = \begin{pmatrix} -1 & -7 & -3 \\ -3 & 3 & 9 \\ -5 & 1 & 3 \end{pmatrix}$$

NT ngược dấu

$$A^{-1} = \frac{1}{\det A} \cdot A^* = \frac{-1}{18} \cdot \begin{pmatrix} -1 & -7 & -3 \\ -3 & 3 & 9 \\ -5 & 1 & 3 \end{pmatrix}$$

Đáp án

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$$X = \begin{pmatrix} 5 & 1 & 3 \\ 2 & 2 & -2 \\ 4 & 2 & 1 \end{pmatrix} \quad \det X = \begin{vmatrix} 5 & 1 & 3 \\ 2 & 2 & -2 \\ 4 & 2 & 1 \end{vmatrix} = -3$$

$$X = \begin{pmatrix} -1 & -7 & -3 \\ -3 & -3 & 9 \\ -5 & 1 & 3 \end{pmatrix} \quad \det X = \begin{vmatrix} -1 & -7 & -3 \\ -3 & -3 & 9 \\ -5 & 1 & 3 \end{vmatrix} = -14$$

$$X = \begin{pmatrix} -1 & -31 & -9 \\ 15 & -27 & 6 \\ -41 & -9 & -14 \end{pmatrix}$$

Đáp án

$$X = \begin{pmatrix} 2 & 1 & -2 \\ 0 & 2 & 1 \\ 3 & -1 & 3 \end{pmatrix} \quad \det X = \begin{vmatrix} 2 & 1 & -2 \\ 0 & 2 & 1 \\ 3 & -1 & 3 \end{vmatrix} = 10 \neq 0$$

$$X = \begin{pmatrix} 2 & 1 & 0 \\ -2 & 1 & 3 \\ 1 & -2 & 5 \end{pmatrix} \quad A^{-1} \oplus$$

Tính A^{-1}

$$\det A = \begin{vmatrix} 2 & 1 & -2 \\ 0 & 2 & 1 \\ 3 & -1 & 3 \end{vmatrix} = 20 \neq 0$$

Tính các phần bù đại số

HÀI TIẾN

$$A_{11} = \begin{vmatrix} 2 & 1 \\ -1 & 3 \end{vmatrix} = 7 \quad A_{21} = - \begin{vmatrix} 1 & -2 \\ -1 & 3 \end{vmatrix} = -$$

$$A_{12} = - \begin{vmatrix} 0 & 1 \\ 3 & 3 \end{vmatrix} = 3 \quad A_{22} = \begin{vmatrix} 2 & -2 \\ 3 & 3 \end{vmatrix} = 12$$

$$A_{13} = \begin{vmatrix} 0 & 2 \\ 3 & -1 \end{vmatrix} = -6 \quad A_{23} = - \begin{vmatrix} 2 & 1 \\ 3 & -1 \end{vmatrix} = 5$$

$$A_{31} = \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} = 5 \quad A_{33} = \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} = 4$$

$$A_{32} = \begin{vmatrix} 2 & -2 \\ 0 & 1 \end{vmatrix} = -2$$

NT phụ hợp:

$$A^* = \begin{pmatrix} 7 & 3 & -6 \\ -1 & 12 & 5 \\ 5 & -2 & 4 \end{pmatrix}^T = \begin{pmatrix} 7 & -1 & 5 \\ 3 & 12 & -2 \\ -6 & 5 & 4 \end{pmatrix}$$

NT nghịch đảo:

$$A^{-1} = \frac{1}{\det A} \cdot A^* = \frac{1}{29} \begin{pmatrix} 7 & -1 & 5 \\ 3 & 12 & -2 \\ -6 & 5 & 4 \end{pmatrix}$$

Tu (*) ta có:

$$X = \begin{pmatrix} 2 & 1 & 0 \\ -2 & 1 & 3 \\ 1 & -2 & 5 \end{pmatrix} \cdot \frac{1}{29} \begin{pmatrix} 7 & -1 & 5 \\ 3 & 12 & -2 \\ -6 & 5 & 4 \end{pmatrix} \quad \text{HÁI TIẾN}$$

$$= \frac{1}{29} \begin{pmatrix} 17 & 10 & 8 \\ -29 & 29 & 0 \\ -29 & 0 & 29 \end{pmatrix}$$

$$\text{bài 1.4.3} \quad \underline{A} = \begin{pmatrix} 4 & 3 \\ 3 & 2 \end{pmatrix} \quad X \begin{pmatrix} 7 & 5 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ -1 & 0 \end{pmatrix}$$

$$X = B^{-1} \begin{pmatrix} 1 & 2 \\ -1 & 0 \end{pmatrix} \cdot A^{-1}$$

• tìm B^{-1} :

$$\det B = \begin{vmatrix} 4 & 3 \\ 3 & 2 \end{vmatrix} = -1 \neq 0 \Rightarrow \exists B^{-1}$$

• Phân bố đảo:

$$A_{11} = 2 \quad A_{21} = -3$$

$$A_{12} = -3 \quad A_{22} = 4$$

• MT phụ hợp:

$$B^* = \begin{pmatrix} 2 & -3 \\ -3 & 4 \end{pmatrix}^T = \begin{pmatrix} 2 & -3 \\ -3 & 4 \end{pmatrix}$$

• MT nghịch đảo:

$$B^{-1} = \frac{1}{\det B} \cdot B^* = - \begin{pmatrix} 2 & -3 \\ -3 & 4 \end{pmatrix} = \begin{pmatrix} -2 & 3 \\ 3 & -4 \end{pmatrix}$$

→ tìm A^{-1} :

$$\det A = \begin{vmatrix} 7 & 5 \\ 3 & 2 \end{vmatrix} = -1 \neq 0 \Rightarrow \exists A^{-1}$$

• Phân bố đảo:

$$A_{11} = 2 \quad A_{21} = -5$$

$$A_{12} = -3 \quad A_{22} = 7$$

• MT phụ hợp:

$$A^* = \begin{pmatrix} 2 & -5 \\ -3 & 7 \end{pmatrix}^T = \begin{pmatrix} 2 & -3 \\ -5 & 7 \end{pmatrix}$$

• MT nghịch đảo:

$$A^{-1} = \frac{1}{\det A} \cdot A^* = -1 \begin{pmatrix} 2 & -3 \\ -5 & 7 \end{pmatrix} = \begin{pmatrix} -2 & 3 \\ 5 & -7 \end{pmatrix} \quad \text{HÁI TIẾN}$$

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1. $X = \begin{pmatrix} -2 & 5 \\ 3 & -4 \end{pmatrix} \begin{pmatrix} 1 & 6 \\ -4 & 0 \end{pmatrix} \begin{pmatrix} -2 & 3 \\ 5 & -2 \end{pmatrix}$

2. $A = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{3 \times 4}$

3. $H_2 - 2H_1 \rightarrow H_2$
 $H_3 - 3H_1 \rightarrow H_3$

4. $\begin{pmatrix} 1 & 1 & 2 & 1 & | & H_2 - H_1 - H_3 \\ 0 & -1 & 0 & 1 \\ 0 & -1 & 0 & 1 \end{pmatrix} \rightarrow \lambda(A) = 2$

5. $\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 4 \\ 3 & 2 & 6 \end{pmatrix} = 0$

6. $\lambda(A) = 2$

7. $A = \begin{pmatrix} 1 & -1 & 3 & 2 & -1 \\ 2 & 2 & -1 & 2 & 3 \\ 1 & -2 & -2 & 2 & -4 \\ 4 & -1 & 0 & 6 & -2 \end{pmatrix}$

8. $\lambda(A) = 3$

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1. $2x_1 + 2x_2 + 5x_3 = 21$
 $2x_1 + 3x_2 + 6x_3 = 26$
 $x_1 - 6x_2 - 9x_3 = -37$

2. $\det A = \begin{vmatrix} 2 & 2 & 5 \\ 2 & 3 & 6 \\ 1 & -6 & -9 \end{vmatrix} = -9$

3. $\det A_1 = \begin{vmatrix} 21 & 2 & 5 \\ 26 & 3 & 6 \\ -37 & -6 & -9 \end{vmatrix} = -102$

4. $\det A_2 = \begin{vmatrix} 2 & 21 & 5 \\ 2 & 26 & 6 \\ 1 & -37 & -9 \end{vmatrix} = -20$

5. $\det A_3 = \begin{vmatrix} 2 & 2 & 21 \\ 2 & 3 & 26 \\ 1 & -6 & -37 \end{vmatrix} = -15$

6. $x_1 = \frac{\det A_1}{\det A} = \frac{4}{3}$
 $x_2 = \frac{\det A_2}{\det A} = \frac{20}{3}$
 $x_3 = \frac{\det A_3}{\det A} = \frac{25}{3}$

7. $x \in \left\{ \frac{4}{3}, \frac{20}{3}, \frac{25}{3} \right\}$

Đáp 2.2 p. Gauss

$$\begin{cases} x_1 + 5x_2 + 2x_3 + 3x_4 - 2x_5 = 4 \\ 4x_1 + x_2 - x_3 + 12x_4 - 8x_5 = 15 \\ 2x_1 + 3x_2 + x_3 + 6x_4 - 4x_5 = 7 \end{cases}$$

Có NT hệ số mở rộng

$$\bar{A} = \left(\begin{array}{cccc|c} 1 & 5 & 2 & 3 & -2 & 4 \\ 4 & 1 & -1 & 12 & -8 & 15 \\ 2 & 3 & 1 & 6 & -4 & 7 \end{array} \right)$$

$H_2 - 4H_1 \rightarrow H_2$
 $H_3 - 2H_1 \rightarrow H_3$

$$\rightarrow \left(\begin{array}{cccc|c} 1 & 5 & 2 & 3 & -2 & 4 \\ 0 & -19 & -9 & 0 & 0 & -11 \\ 0 & -7 & -3 & 0 & 0 & -1 \end{array} \right)$$

$H_1 \leftrightarrow H_2$
 $H_2 \leftrightarrow H_3$

$$\rightarrow \left(\begin{array}{cccc|c} 1 & -7 & -3 & 0 & 0 & -1 \\ 0 & -19 & -9 & 0 & 0 & -11 \\ 0 & -1 & -1 & 0 & 0 & -1 \end{array} \right)$$

$H_1 + 7H_2 \rightarrow H_1$
 $H_2 + 19H_3 \rightarrow H_2$

$$\rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 & -8 \\ 0 & -1 & -1 & 0 & 0 & -1 \\ 0 & -1 & -1 & 0 & 0 & -1 \end{array} \right)$$

$H_1 - H_2 \rightarrow H_1$
 $H_3 - H_2 \rightarrow H_3$

$$\rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 0 & -7 \\ 0 & -1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$\lambda(A) = \lambda(\bar{A}) = 3 < 5 \rightarrow$ hệ có vô số nghiệm

hệ tương đương

$$\begin{cases} x_1 + 5x_2 + 2x_3 + 3x_4 - 2x_5 = 4 \\ -x_2 - x_3 = -1 \\ 6x_3 = -12 \end{cases}$$

$x_3 = -2$
 $x_2 = 1$
 x_4, x_5 tùy ý

Đáp 2.3 p. Gauss

$$\begin{cases} 2x_1 - 3x_2 + 3x_3 - 2x_4 + x_5 = 14 \\ 5x_1 + 2x_2 + 3x_3 - 2x_4 - 6x_5 = 17 \\ 3x_1 + 2x_2 + x_3 + 2x_4 - 8x_5 = -1 \end{cases}$$

Có NT hệ số mở rộng

$$\bar{A} = \left(\begin{array}{ccccc|c} 2 & -3 & 3 & -2 & 1 & 14 \\ 5 & 2 & 3 & -2 & -6 & 17 \\ 3 & 2 & 1 & 2 & -8 & -1 \end{array} \right)$$

$H_2 - 5H_1 \rightarrow H_2$
 $H_3 - 3H_1 \rightarrow H_3$

$$\rightarrow \left(\begin{array}{ccccc|c} 2 & -3 & 3 & -2 & 1 & 14 \\ 0 & 17 & -12 & 8 & -11 & -59 \\ 0 & 11 & -8 & 8 & -11 & -23 \end{array} \right)$$

$H_2 \leftrightarrow H_3$
 $H_1 \leftrightarrow H_2$

$$\rightarrow \left(\begin{array}{ccccc|c} 0 & 11 & -8 & 8 & -11 & -23 \\ 2 & -3 & 3 & -2 & 1 & 14 \\ 0 & 17 & -12 & 8 & -11 & -59 \end{array} \right)$$

$H_1 - 2H_2 \rightarrow H_1$
 $H_3 - 17H_2 \rightarrow H_3$

$$\rightarrow \left(\begin{array}{ccccc|c} 0 & 11 & -8 & 8 & -11 & -23 \\ 2 & -3 & 3 & -2 & 1 & 14 \\ 0 & 0 & 80 & -64 & 8 & -35 \end{array} \right)$$

$H_1 + 3H_2 \rightarrow H_1$
 $H_3 - 8H_2 \rightarrow H_3$

$$\rightarrow \left(\begin{array}{ccccc|c} 0 & 11 & -8 & 8 & -11 & -23 \\ 2 & 9 & -21 & 14 & 2 & 56 \\ 0 & 0 & 80 & -64 & 8 & -35 \end{array} \right)$$

$H_1 - 9H_2 \rightarrow H_1$
 $H_3 - 8H_2 \rightarrow H_3$

$$\rightarrow \left(\begin{array}{ccccc|c} 0 & 11 & -8 & 8 & -11 & -23 \\ 0 & 0 & 80 & -64 & 8 & -35 \\ 0 & 0 & 80 & -64 & 8 & -35 \end{array} \right)$$

$H_3 - H_2 \rightarrow H_3$

$$\rightarrow \left(\begin{array}{ccccc|c} 0 & 11 & -8 & 8 & -11 & -23 \\ 0 & 0 & 80 & -64 & 8 & -35 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$\lambda(A) = \lambda(\bar{A}) = 3 < 5 \rightarrow$ hệ có vô số nghiệm

hệ tương đương

$$\begin{cases} 11x_2 - 8x_3 + 8x_4 - 11x_5 = -23 \\ 80x_3 - 64x_4 + 8x_5 = -35 \end{cases}$$

$x_3 = -\frac{35}{80} = -\frac{7}{16}$
 $x_2 = \frac{1}{11}$
 x_4, x_5 tùy ý

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H.T. mđ. long

$$A = \begin{pmatrix} 1 & 2 & -2 & 1 & 3 \\ 2 & 3 & 1 & -2 & 4 \\ 3 & 5 & -2 & 2 & 6 \\ 6 & 10 & -3 & 1 & 13 \end{pmatrix}$$

$H_2 = 2H_1 \rightarrow H_2$
 $H_3 = 3H_1 \rightarrow H_3$
 $H_4 = 6H_1 \rightarrow H_4$

$$\begin{pmatrix} 1 & 2 & -2 & 1 & 3 \\ 0 & -1 & 5 & -4 & -2 \\ 0 & -1 & 4 & -1 & -3 \\ 0 & -2 & 9 & -5 & -5 \end{pmatrix}$$

$H_2 - H_3 \rightarrow H_2$
 $H_4 - 2H_3 \rightarrow H_4$

$$\begin{pmatrix} 1 & 2 & -2 & 1 & 3 \\ 0 & -1 & 5 & -4 & -2 \\ 0 & 0 & -1 & 3 & -1 \\ 0 & 0 & -1 & 3 & -1 \end{pmatrix}$$

$H_4 - H_3 \rightarrow H_4$

$$\begin{pmatrix} 1 & 2 & -2 & 1 & 3 \\ 0 & -1 & 5 & -4 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$\lambda(A) = \lambda(B) = 3 < 4$ (mđ. long)
 $\Rightarrow$ mđ. long mđ.

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Memo No. _____
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H.T. mđ. long

$$A = \begin{pmatrix} 1 & 2 & -2 & 1 & 3 \\ 2 & 3 & 1 & -2 & 4 \\ 3 & 5 & -2 & 2 & 6 \\ 6 & 10 & -3 & 1 & 13 \end{pmatrix}$$

$H_2 = 2H_1 \rightarrow H_2$
 $H_3 = 3H_1 \rightarrow H_3$
 $H_4 = 6H_1 \rightarrow H_4$

$$\begin{pmatrix} 1 & 2 & -2 & 1 & 3 \\ 0 & -1 & 5 & -4 & -2 \\ 0 & -1 & 4 & -1 & -3 \\ 0 & -2 & 9 & -5 & -5 \end{pmatrix}$$

$H_2 - H_3 \rightarrow H_2$
 $H_4 - 2H_3 \rightarrow H_4$

$$\begin{pmatrix} 1 & 2 & -2 & 1 & 3 \\ 0 & -1 & 5 & -4 & -2 \\ 0 & 0 & -1 & 3 & -1 \\ 0 & 0 & -1 & 3 & -1 \end{pmatrix}$$

$H_4 - H_3 \rightarrow H_4$

$$\begin{pmatrix} 1 & 2 & -2 & 1 & 3 \\ 0 & -1 & 5 & -4 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$\lambda(A) = \lambda(B) = 3 < 4$ (mđ. long)
 $\Rightarrow$ mđ. long mđ.

HÀI TIẾN

$$\det A_3 = \begin{vmatrix} 1 & 2 & 6 & 2 \\ 2 & 3 & 7 & 1 \\ 3 & 5 & 2 & 4 \\ 4 & 6 & 2 & 2 \end{vmatrix} = -16$$

$$\det A_4 = \begin{vmatrix} 1 & 2 & -3 & 6 \\ 2 & 3 & 1 & 7 \\ 3 & 5 & 2 & 23 \\ 4 & 6 & 6 & 22 \end{vmatrix} = 8$$

$$\begin{cases} x_1 = \det A_1 / \det A = -64 \\ x_2 = 43 \\ x_3 = 4 \\ x_4 = -2 \end{cases}$$

Hay re cuoi ket pt $x \in \{-64, 43, 4, -2\}$
Do tuip.

Cho $A = \begin{pmatrix} 5 & -1 & -1 \\ 1 & 2 & 3 \end{pmatrix}$ $B = \begin{pmatrix} 3 & 2 \\ 2 & 1 \\ 4 & 3 \end{pmatrix}$

a) Tinh $\det(A \cdot B)$, $\det(B \cdot A)$

b) Tinh rang $BA + SI$

a) $A \cdot B = \begin{pmatrix} 3 & -1 & -1 \\ 1 & 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 \\ 2 & 1 \\ 4 & 3 \end{pmatrix} = A \cdot B$

HAI TIEN

$$= \begin{pmatrix} 7 & 4 \\ 19 & 13 \end{pmatrix}$$

$$\det(A \cdot B) = \begin{vmatrix} 7 & 4 \\ 19 & 13 \end{vmatrix} = 7 \cdot 13 - 19 \cdot 4 = 15$$

$$B \cdot A = \begin{pmatrix} 3 & 2 \\ 2 & 1 \\ 4 & 3 \end{pmatrix} \cdot \begin{pmatrix} 3 & 1 & -1 \\ 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 11 & 7 & 3 \\ 7 & 4 & 1 \\ 15 & 10 & 5 \end{pmatrix}$$

$$\det(B \cdot A) = \begin{vmatrix} 11 & 7 & 3 \\ 7 & 4 & 1 \\ 15 & 10 & 5 \end{vmatrix} = 11 \cdot 4 \cdot 5 + 7 \cdot 1 \cdot 15 + 7 \cdot 10 \cdot 3 - 15 \cdot 4 \cdot 3 - 7 \cdot 7 \cdot 5 - 10 \cdot 1 \cdot 11 = 0$$

b) $BA + SI = \begin{pmatrix} 11 & 7 & 3 \\ 7 & 4 & 1 \\ 15 & 10 & 5 \end{pmatrix} + 5 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 16 & 7 & 3 \\ 7 & 9 & 1 \\ 15 & 10 & 10 \end{pmatrix}$$

$\det(BA + SI) = \begin{vmatrix} 16 & 7 & 3 \\ 7 & 9 & 1 \\ 15 & 10 & 10 \end{vmatrix} = 16 \cdot 9 \cdot 10 + 7 \cdot 10 \cdot 3 + 7 \cdot 1 \cdot 15 - 15 \cdot 9 \cdot 1 - 7 \cdot 7 \cdot 10 - 10 \cdot 10 \cdot 1 = 700 \neq 0$

HAI TIEN

$$A = \begin{pmatrix} -3 & 1 & 1 \\ 3 & 3 & -1 \\ 4 & 1 & 5 \end{pmatrix} \quad B = \begin{pmatrix} 12 & 1 & 1 \\ 1 & -2 & 2 \\ 2 & 0 & 2 \end{pmatrix}$$

$$a) \quad \det(A) = \begin{vmatrix} -3 & 1 & 1 \\ 3 & 3 & -1 \\ 4 & 1 & 5 \end{vmatrix} = (-3)(15 - 1) - 1(15 - 4) + 4(15 - 12) = -36 - 11 + 12 = -35$$

$$b) \quad \det(B) = \begin{vmatrix} 12 & 1 & 1 \\ 1 & -2 & 2 \\ 2 & 0 & 2 \end{vmatrix} = 12(4 - 0) - 1(2 - 4) + 1(2 - 4) = 48 - 2 - 2 = 44$$

$$\det(AB) = \det(A) \cdot \det(B) = (-35) \cdot 44 = -1540$$

$$c) \quad \det(A+B) = \begin{vmatrix} 9 & 4 & 2 \\ 6 & 6 & 0 \\ 8 & 2 & 10 \end{vmatrix} = 9(60 - 0) - 4(60 - 0) + 2(60 - 48) = 540 - 240 + 120 = 420$$

$$d) \quad \det(A-B) = \begin{vmatrix} -14 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

$$e) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$f) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$g) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$h) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$i) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$j) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$k) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$l) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$m) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$n) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$o) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$p) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$q) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$r) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$s) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$t) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$u) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$v) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$w) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$x) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$y) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$z) \quad \det(A) = -35, \det(B) = 44, \det(A+B) = 420, \det(A-B) = 0$$

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 2 \\ 2 & 0 & 2 \end{pmatrix} \quad B = \begin{pmatrix} -3 & 2 & 1 \\ 3 & 3 & -1 \\ 4 & 1 & 5 \end{pmatrix}$$

$$a) \quad \det(A) = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 2 \\ 2 & 0 & 2 \end{vmatrix} = 2(10 - 2) - 1(2 - 4) + 1(2 - 4) = 16 - 2 - 2 = 12$$

$$b) \quad \det(B) = \begin{vmatrix} -3 & 2 & 1 \\ 3 & 3 & -1 \\ 4 & 1 & 5 \end{vmatrix} = -3(15 - 5) - 2(15 - 20) + 1(15 - 12) = -30 + 10 + 3 = -17$$

$$c) \quad \det(A+B) = \begin{vmatrix} -1 & 3 & 2 \\ 4 & 5 & 4 \\ 6 & 1 & 7 \end{vmatrix} = -1(35 - 28) - 3(28 - 28) + 2(28 - 30) = -7 - 0 - 4 = -11$$

$$d) \quad \det(A-B) = \begin{vmatrix} 5 & -1 & 0 \\ -2 & -1 & 0 \\ 0 & 2 & -3 \end{vmatrix} = 5(3 - 0) - (-1)(6 - 0) + 0(6 - 0) = 15 + 6 = 21$$

$$e) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$f) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$g) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$h) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$i) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$j) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$k) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$l) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$m) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$n) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$o) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$p) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$q) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$r) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$s) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$t) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$u) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$v) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$w) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$x) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$y) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

$$z) \quad \det(A) = 12, \det(B) = -17, \det(A+B) = -11, \det(A-B) = 21$$

ĐB' là d. 10. ch (e)
 e) $\lambda = 10 \neq 0$
 b) $\lambda \neq 5$
 $\left. \begin{array}{l} 2x_1 + x_2 - x_3 = 6 \\ 3x_1 + x_2 + 4x_3 = 0 \\ 2x_1 + 4x_2 + 4x_3 = 2 \end{array} \right\}$
 $A = \begin{pmatrix} 2 & 3 & -1 \\ 3 & 1 & 4 \\ 2 & 4 & 4 \end{pmatrix} \left| \begin{array}{c} 6 \\ 0 \\ 2 \end{array} \right|$

$2H_2 - H_1 \rightarrow H_3$
 $H_3 - H_1 \rightarrow H_3$
 $-H_3 - H_1 \rightarrow H_3$

$\lambda(A) = \lambda(A) = 3 \rightarrow$ kđ pt có 3 nghiệm khác nhau

Hệ pt tương đương
 $\left. \begin{array}{l} 2x_1 + 3x_2 - x_3 = 6 \\ -7x_2 + 4x_3 = -18 \end{array} \right\} \begin{array}{l} x_1 = 9/2 \\ x_2 = 29/19 \\ x_3 = -46/19 \end{array}$

$10 \neq 2.8$

$\left. \begin{array}{l} 2x_1 + 3x_2 - x_3 + 2x_4 = 6 \\ 10x_1 + x_2 + 5x_3 + x_4 = 9 \\ 3x_1 + 5x_2 - 5x_3 + (4+5)x_4 = 3 \end{array} \right\}$

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$A = \begin{pmatrix} 2 & 3 & -1 & 2 \\ 1 & 1 & 3 & 1 \\ 3 & 5 & -5 & \alpha+5 \end{pmatrix} \left| \begin{array}{c} 6 \\ 9 \\ 3 \end{array} \right|$

$2H_2 - H_1 \rightarrow H_3$
 $2H_3 - 3H_1 \rightarrow H_3$

$H_3 + H_2 \rightarrow H_3$

$\left(\begin{array}{cccc|c} 2 & 3 & -1 & 2 & 6 \\ 0 & -1 & 7 & 0 & 12 \\ 0 & 0 & 0 & 2\alpha+4 & 0 \end{array} \right)$

TH1: Nếu $2\alpha+4 = 0 \Leftrightarrow \alpha = -2$
 thì $\lambda(A) = \lambda(A) = 2 < 4 \rightarrow$ Hệ có vô số nghiệm

Hệ tương đương
 $\left. \begin{array}{l} 2x_1 + 3x_2 - x_3 + 2x_4 = 6 \\ -x_2 + 7x_3 = 12 \end{array} \right\} \begin{array}{l} x_1 = 21 - 10x_3 - 2x_4 \\ x_2 = 7x_3 - 12 \end{array}$

TH2: Nếu $2\alpha+4 \neq 0 \Leftrightarrow \alpha \neq -2$
 thì $\lambda(A) \neq \lambda(A) = 2 < 4 \rightarrow$ Hệ vô nghiệm

Hệ tương đương
 $\left. \begin{array}{l} 2x_1 + 3x_2 - x_3 + 2x_4 = 6 \\ -x_2 + 7x_3 = 12 \\ (2\alpha+4)x_4 = 0 \end{array} \right\}$

e) $\left. \begin{array}{l} x_1 = 21 - 10x_3 - 2x_4 \\ x_2 = 7x_3 - 12 \\ x_4 = 0 \end{array} \right\}$

Hải Tiến

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$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Hier by day: $\begin{cases} 3x_1 = 3 \\ 2x_1 + 5x_2 = 1 \end{cases} \Leftrightarrow \begin{cases} x_1 = 1 \\ x_2 = -1 \end{cases}$
 Varg cd below: $= 2a_1 + a_2 = a$

Đuối tay: $\text{Tiên} \times \text{Đuối} \rightarrow A \text{ thu} \text{ righ} \text{t} \text{B}$

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Def: A ist nglisch $\Leftrightarrow \det(A) \neq 0$

42 HẢI TIỀN

Mon	Tue	Wed	Thurs	Fri	Sat	Sun
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6700
thru C.C.

$C_1 = 1.8 \times 10^6$

$$\frac{C_1 \lambda^k (1-\lambda)^k}{(k-1)!} \begin{array}{c|ccc} & \lambda & 1-\lambda & \\ \hline \lambda & \lambda & 0 & \\ 1-\lambda & \lambda & 1-\lambda & \end{array}$$

$$\frac{1}{n} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x & x & x \\ x & x & x \\ x & x & x \end{bmatrix}$$

$$+ (n-1)(1-x) \begin{bmatrix} x & x_1 \\ x & x \end{bmatrix}$$

$$\frac{(x-1)(x-1) - (x-1)(x-x^2)}{(x-1)(x-1)}$$

$$+ (x-1)(1-x)(x-x^2)$$

$$= \frac{(x-1)(x-1) \cdots (x-1)^{n-1}}{(x-1 - x + x^2) \cdots (x-1 - x + x^2)^{n-1}} (x - x^c)$$

$$= -(x-1) \left[(x-1)(x^2-1) + (1-x)(x-x^2) \right]$$

$$= (x-1)^2 (x^2 - 1 - (x+x^2)) = (x-1)^2 (2x^2 - x - 1)$$

$$\det A \neq 0 \Leftrightarrow \begin{cases} (x-1)^L \neq 0 & \text{e) } x \neq 1 \\ 2x^L - x - 1 \neq 0 & x \neq \frac{1}{2} \end{cases}$$

HẢI TẤN

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1. $A^{-1} = \frac{1}{\det A} A^* = \frac{1}{1} \begin{pmatrix} 1 & 1 & -1 \\ -9 & -2 & 3 \\ 15 & 8 & -11 \end{pmatrix}$

b) $XA = B$
 e) $X = B A^{-1}$

$$= \begin{pmatrix} 5 & 2 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & -1 \\ -9 & -2 & 3 \\ 15 & 8 & -11 \end{pmatrix}$$

$$= \begin{pmatrix} 10 & 7 & -8 \\ 12 & 7 & -9 \\ 5 & 3 & -9 \end{pmatrix}$$

3. $A = \begin{pmatrix} -2 & 1 & 3 \\ -1 & 1 & -4 \\ -3 & 2 & -2 \end{pmatrix}, B = \begin{pmatrix} 3 & 2 & -2 \\ 1 & 2 & 4 \end{pmatrix}$

a) Tìm A^{-1}
 $\det A = \begin{vmatrix} -2 & 1 & 3 \\ -1 & 1 & -4 \\ -3 & 2 & -2 \end{vmatrix} = 1 \neq 0$
 Tìm phần bù đại số:

$$A_{11} = \begin{vmatrix} 1 & -4 \\ 2 & -2 \end{vmatrix} = 6$$

$$A_{22} = \begin{vmatrix} -2 & 3 \\ -3 & -2 \end{vmatrix} = 8$$

HÀI TIẾN

Memo No. _____ Date _____

$A_{12} = \begin{vmatrix} -1 & -9 \\ -3 & -2 \end{vmatrix} = 10$

$$A_{23} = \begin{vmatrix} -2 & 3 \\ -3 & -2 \end{vmatrix} = -13$$

$A_{43} = \begin{vmatrix} -1 & 1 \\ -3 & 2 \end{vmatrix} = 1$

$$A_{21} = \begin{vmatrix} -2 & 1 \\ -3 & 2 \end{vmatrix} = 1$$

$A_{11} = \begin{vmatrix} 1 & 3 \\ 1 & -4 \end{vmatrix} = -7$

$$A_{23} = \begin{vmatrix} -2 & 1 \\ -3 & 2 \end{vmatrix} = -1$$

$A_{32} = \begin{vmatrix} -1 & 3 \\ 1 & -4 \end{vmatrix} = -11$

NT. Phần bù

$$A^* = \begin{pmatrix} 6 & 10 & 1 \\ 8 & 13 & -11 \\ -7 & -11 & -1 \end{pmatrix}$$

NT. nghịch đảo

$$A^{-1} = \frac{1}{\det A} A^* = \frac{1}{1} \begin{pmatrix} 6 & 8 & -7 \\ 10 & 13 & -11 \\ 1 & 1 & -1 \end{pmatrix}$$

b) $XA = B$ e) $X = A^{-1} B A^{-1}$

$$X = \begin{pmatrix} 3 & 2 & -2 \\ 1 & 2 & 4 \end{pmatrix} \begin{pmatrix} 6 & 8 & -7 \\ 10 & 13 & -11 \\ 1 & 1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 36 & 48 & -41 \\ 30 & 38 & -38 \end{pmatrix}$$

HÀI TIẾN

Đang: $(\text{Cao} \times H)$
 p. chung: Đià vớ NT Affany

Bài 1.2.2 (cố chứa)
 $2H_i + 2H_j \times H_i$ e. riêng

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & x & x & 1 & 1 \\ x & x & 1 & 1 & 1 \\ x & 1 & x & 1 & 1 \\ 1 & 1 & 1 & 1 & x \end{pmatrix} \xrightarrow{H_2-H_1 \rightarrow H_2} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & x-1 & x-1 & 0 & 0 \\ 0 & 0 & 1-x & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & x-1 \end{pmatrix}$$

$$H_4 + H_2 \rightarrow H_4 \rightarrow \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & x-1 & x-1 & 1-x & 1-x \\ 0 & 0 & 1-x & 1-x & 1-x \\ 0 & 0 & 0 & 1-x & 1-x \\ 0 & 0 & 0 & 0 & 2-x-x^2 \end{pmatrix} \xrightarrow{H_3-H_2 \rightarrow H_3} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & x-1 & x-1 & 1-x & 1-x \\ 0 & 0 & 1-x & 1-x & 1-x \\ 0 & 0 & 0 & 1-x & 1-x \\ 0 & 0 & 0 & 0 & 2-x-x^2 \end{pmatrix} = B$$

• Nếu: $3-x-2x^2 = 0 \Leftrightarrow x = -\frac{3}{2}$

$$B = \begin{pmatrix} 1 & 1 & 1 & -3/2 \\ 0 & -5/2 & -5/2 & 5/2 \\ 0 & 0 & 5/2 & -5/2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow \lambda(A) = \lambda(B) = 3$

HÀI TIỀN

• Nếu: $x = 1$ thì $B = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \lambda(A) = \lambda(B) = 1$

Nếu: $x \neq 1$ thì $\lambda(A) = 4$
 $x \neq -3/2$

Bài 1.2.8

$$A = \begin{pmatrix} 1 & x & 1 & x \\ 1 & x & x & x \\ x & 1 & 1 & x \\ x & 1 & 1 & 1 \end{pmatrix} \xrightarrow{H_2-H_1 \rightarrow H_2} \begin{pmatrix} 1 & x & 1 & x \\ 0 & 0 & x-1 & 0 \\ 0 & 1-x^2 & 1-x & x-x^2 \\ 0 & 1-x^2 & 1-x & 1-x^2 \end{pmatrix}$$

$C \Leftrightarrow G_3 \rightarrow \begin{pmatrix} 1 & 1 & x & x \\ 0 & x-1 & 0 & 0 \\ 0 & 1-x & 1-x^2 & x-x^2 \\ 0 & 1-x & 1-x^2 & 1-x^2 \end{pmatrix}$

$H_3 + H_2 \rightarrow H_3 \rightarrow \begin{pmatrix} 1 & 1 & x & x \\ 0 & x-1 & 0 & 0 \\ 0 & 0 & 1-x^2 & x-x^2 \\ 0 & 0 & 1-x^2 & 1-x^2 \end{pmatrix}$

$H_4 + H_2 \rightarrow H_4 \rightarrow \begin{pmatrix} 1 & 1 & x & x \\ 0 & x-1 & 0 & 0 \\ 0 & 0 & 1-x^2 & x-x^2 \\ 0 & 0 & 1-x^2 & 1-x^2 \end{pmatrix} = B$

Nếu: $x = 1$ thì $\lambda(A) = 1$
 Nếu: $x = -1$ thì $\lambda(A) = 3$
 Nếu: $x \neq 1$ thì $\lambda(A) = 4$
 $x \neq -1$

HÀI TIỀN

$$\begin{cases} 3x_1 + x_2 + x_3 - x_4 - x_5 = 3 \\ 2x_1 + 5x_2 - 2x_3 + 4x_4 + x_5 = 7 \\ 3x_1 + 4x_2 - 2x_3 + x_4 - 2x_5 = 4 \\ 6x_1 + 8x_2 - 3x_3 + 4x_4 - 2x_5 = \lambda \end{cases}$$

Hãy pt mđ xog.

$$A = \begin{pmatrix} 1 & 1 & 1 & -1 & -1 \\ 2 & 3 & -2 & 4 & 1 \\ 3 & 4 & -2 & 1 & -2 \\ 6 & 8 & -3 & 4 & -2 \end{pmatrix} \begin{vmatrix} 3 \\ 7 \\ 4 \\ \lambda \end{vmatrix}$$

$$\begin{aligned} & 2H_1 \rightarrow H_2 \\ & 3H_1 \rightarrow H_3 \\ & 6H_1 \rightarrow H_4 \\ & \rightarrow H_2 \begin{pmatrix} 1 & 1 & 1 & -1 & -1 \\ 0 & 1 & -4 & 6 & 3 \\ 0 & 1 & -5 & 4 & 1 \\ 0 & 2 & -13 & 10 & 4 \end{pmatrix} \begin{vmatrix} 3 \\ 3 \\ 1 \\ \lambda-18 \end{vmatrix} \\ & \rightarrow H_3 \begin{pmatrix} 1 & 1 & 1 & -1 & -1 \\ 0 & 1 & -4 & 6 & 3 \\ 0 & 0 & -1 & -2 & -8 \\ 0 & 0 & -4 & -2 & -20 \end{pmatrix} \begin{vmatrix} 3 \\ 3 \\ 1 \\ \lambda-20 \end{vmatrix} \\ & \rightarrow H_4 \begin{pmatrix} 1 & 1 & 1 & -1 & -1 \\ 0 & 1 & -4 & 6 & 3 \\ 0 & 0 & -1 & -2 & -8 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{vmatrix} 3 \\ 3 \\ 1 \\ -4 \end{vmatrix} \end{aligned}$$

$$\begin{cases} x_1 = -28 + 17x_4 + 15x_5 \\ x_2 = 25 - 14x_4 - 11x_5 \\ x_3 = 6 - 2x_4 - 2x_5 \\ x_4, x_5 \text{ tự g} \end{cases}$$

HÀI TIỀN

$$\begin{cases} 3x_1 - 2x_2 + x_3 - 2x_4 = 4 \\ 2x_1 - x_2 + 3x_3 + 3x_4 = 3 \\ 4x_1 - 3x_2 - x_3 - 7x_4 = \lambda \end{cases}$$

$$A = \begin{pmatrix} 3 & -2 & 1 & -2 \\ 2 & -1 & 3 & 3 \\ 4 & -3 & -1 & -7 \end{pmatrix} \begin{vmatrix} 4 \\ 3 \\ \lambda \end{vmatrix}$$

$$\begin{aligned} & 3H_1 - 2H_2 \rightarrow H_3 \\ & 3H_2 - 4H_1 \rightarrow H_3 \\ & H_2 + H_3 \rightarrow H_3 \\ & \rightarrow H_3 \begin{pmatrix} 3 & -2 & 1 & -2 \\ 0 & 1 & 7 & 13 \\ 0 & 1 & 7 & 13 \end{pmatrix} \begin{vmatrix} 4 \\ 3 \\ 3\lambda-15 \end{vmatrix} \\ & \rightarrow H_3 \begin{pmatrix} 3 & -2 & 1 & -2 \\ 0 & 1 & 7 & 13 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{vmatrix} 4 \\ 3 \\ 3\lambda-15 \end{vmatrix} \\ & \rightarrow H_3 \begin{pmatrix} 3 & -2 & 1 & -2 \\ 0 & 1 & 7 & 13 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{vmatrix} 4 \\ 3 \\ 3\lambda-15 \end{vmatrix} \end{aligned}$$

$$\begin{cases} 3x_1 - 2x_2 + x_3 - 2x_4 = 0 \\ 2x_1 - x_2 + 3x_3 + 3x_4 = 0 \\ 4x_1 - 3x_2 - x_3 - 7x_4 = 0 \end{cases}$$

$$A = \begin{pmatrix} 3 & -2 & 1 & -2 \\ 2 & -1 & 3 & 3 \\ 4 & -3 & -1 & -7 \end{pmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix}$$

$$\begin{aligned} & 3H_1 - 2H_2 \rightarrow H_3 \\ & 3H_2 - 4H_1 \rightarrow H_3 \\ & H_2 + H_3 \rightarrow H_3 \\ & \rightarrow H_3 \begin{pmatrix} 3 & -2 & 1 & -2 \\ 0 & 1 & 7 & 13 \\ 0 & 1 & 7 & 13 \end{pmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix} \\ & \rightarrow H_3 \begin{pmatrix} 3 & -2 & 1 & -2 \\ 0 & 1 & 7 & 13 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix} \end{aligned}$$

HÀI TIỀN

$$A = \begin{pmatrix} 1 & -1 & 1 & 1 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 2 & 2 \end{pmatrix} \quad \left| \begin{array}{c} H_1 - H_2 \rightarrow H_3 \\ H_4 - H_3 \rightarrow H_4 \end{array} \right.$$

$$A = \begin{pmatrix} 1 & -1 & 1 & 1 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 4 \end{pmatrix}$$

Hệ pt tương đương:

$$\begin{cases} x_1 - x_2 + x_3 + x_4 = 0 \\ 2x_2 - 2x_3 = 0 \\ 2x_3 - 2x_4 = 0 \\ 4x_4 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = -x_2 + x_3 + x_4 \\ x_2 = x_3 \\ x_3 = x_4 \\ x_4 = 0 \end{cases}$$

Vậy $x \in \left\{ \begin{pmatrix} -x_2 + x_3 + x_4 \\ x_3 \\ x_3 \\ 0 \end{pmatrix} \right\}$

$$A = \begin{pmatrix} 1 & 1 & -3 & -3 \\ 0 & 1 & 10 & 5 \\ 0 & 1 & 11 & 11 \\ 0 & 2 & 22 & 22 \end{pmatrix} \quad \left| \begin{array}{c} H_2 - H_1 \rightarrow H_2 \\ H_3 - H_1 \rightarrow H_3 \\ H_4 - 2H_1 \rightarrow H_4 \end{array} \right.$$

$$A = \begin{pmatrix} 1 & 1 & -3 & -3 \\ 0 & 1 & 10 & 5 \\ 0 & 1 & 11 & 11 \\ 0 & 0 & 2 & 12 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 & -3 & -3 \\ 0 & 1 & 10 & 5 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 2 & 12 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 & -3 & -3 \\ 0 & 1 & 10 & 5 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Hệ pt tương đương:

$$\begin{cases} x_1 + x_2 - 3x_3 - 3x_4 = 0 \\ x_2 + 10x_3 + 5x_4 = 0 \\ x_3 + 6x_4 = 0 \\ 0 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = -x_2 + 3x_3 + 3x_4 \\ x_2 = -10x_3 - 5x_4 \\ x_3 = -6x_4 \end{cases}$$

Vậy $x \in \left\{ \begin{pmatrix} -x_2 + 3x_3 + 3x_4 \\ -10x_3 - 5x_4 \\ -6x_4 \end{pmatrix} \right\}$

$$\begin{cases} x_1 + x_2 - 3x_3 - 5x_4 = 5 \\ x_2 + 10x_3 + 5x_4 = -1 \\ x_3 + 6x_4 = 0 \end{cases}$$

$$\begin{cases} x_1 = 4 - 10x_3 \\ x_2 = -1 + 55x_3 \\ x_3 = -6x_4 \end{cases}$$

Vậy $x \in \{ 4 - 10x_3, -1 + 55x_3, -6x_4, x_4 \}$
 x_4 tùy ý

Bài tập chương III

Bài 3.2

Giải hệ phương trình $A_1x + A_2x + A_3x = b$

$$\Rightarrow A_1 \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} + A_2 \begin{pmatrix} 3 \\ -9 \\ 1 \end{pmatrix} + A_3 \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ -6 \\ 1 \end{pmatrix}$$

$$\begin{cases} \lambda_1 + 3\lambda_2 - 3\lambda_3 = 5 \\ \lambda_1 - 9\lambda_2 + 2\lambda_3 = -6 \\ -2\lambda_1 + \lambda_2 + \lambda_3 = 1 \end{cases}$$

$$A = \begin{pmatrix} 1 & 3 & -3 \\ 1 & -9 & 2 \\ -2 & 1 & 1 \end{pmatrix} \xrightarrow{H_2 - H_1} \begin{pmatrix} 1 & 3 & -3 \\ 0 & -10 & 5 \\ -2 & 1 & 1 \end{pmatrix} \xrightarrow{H_3 + 2H_2} \begin{pmatrix} 1 & 3 & -3 \\ 0 & -10 & 5 \\ 0 & -19 & 11 \end{pmatrix}$$

$$\xrightarrow{H_2 \div (-10)} \begin{pmatrix} 1 & 3 & -3 \\ 0 & 1 & -0.5 \\ 0 & -19 & 11 \end{pmatrix} \xrightarrow{H_3 + 19H_2} \begin{pmatrix} 1 & 3 & -3 \\ 0 & 1 & -0.5 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} \lambda_1 + 3\lambda_2 - 3\lambda_3 = 5 \\ -\lambda_2 + 5\lambda_3 = -11 \end{cases}$$

$$\begin{cases} \lambda_1 = (6\lambda_3 + 2)/7 \\ \lambda_2 = (5\lambda_3 + 11)/7 \end{cases}$$

λ_3 tùy ý

Vậy ta có đáp án:

$$x = \frac{6\lambda_3 + 2}{7} q_1 + \frac{5\lambda_3 + 11}{7} q_2 + \lambda_3 q_3$$

HAY TÍNH

HAY TÍNH